

STA304 — Learning Activity 7 Solutions

Cluster Sampling with Equal Probability

Summer 2026

Exercise 1

An inspector samples cans from a truckload of canned creamed corn to estimate the total number of worm fragments. The truck has 580 cases; each case contains 24 cans. The inspector samples 12 cases at random, and subsamples 3 cans randomly from each selected case.

Part (a): One-stage or two-stage cluster sampling?

This is a **two-stage cluster sampling** design because we perform a SRS on selecting the number of cans (SRS on SSUs) *after* randomly selecting the 12 cases (PSUs). In other words, we randomly sample the cases at the first stage and randomly sample cans within each selected case at the second stage. It would be one-stage cluster sampling only if we sampled the cases (first stage) and then observed *all* 24 cans in each selected case, leaving no second stage of random sampling.

Part (b): PSUs

The **PSUs are the cases** (boxes containing the cans).

$$N = \text{TOTAL CASES} = \text{population number of PSUs} = 580$$

$$n = \text{SAMPLED CASES} = \text{sampled number of PSUs} = 12$$

Part (c): Estimate the total number of worm fragments

We know that $N = 580$, $n = 12$, $M_i = M = 24$, and $m_i = m = 3$.

Estimated PSU total. For each sampled case i , the estimated total is:

$$\hat{t}_i = \frac{M_i}{m_i} \sum_j y_{ij} = \frac{24}{3} \sum_j y_{ij} = 8 \sum_j y_{ij}$$

Summing over sampled cases:

$$\sum_{i \in S} \hat{t}_i = 1048 \quad \implies \quad \bar{t}_S = \frac{1048}{12} = 87.3333$$

Estimated population total:

$$\hat{t} = N\bar{t}_S = 580 \times 87.3333 = \mathbf{50,653.3333}$$

Standard error. First compute the two variance components needed:

$$s_t^2 = \frac{1}{n-1} \sum_{i \in S} (\hat{t}_i - \bar{t}_S)^2 = 2611.8788$$

$$\sum_{i \in S} s_i^2 = 54.3333$$

where s_i^2 is the sample variance of the $m = 3$ observed worm counts within case i .

Plugging into the estimated variance formula:

$$\begin{aligned} \hat{V}(\hat{t}) &= N^2 \left(1 - \frac{n}{N}\right) \frac{s_t^2}{n} + \frac{N}{n} \sum_{i \in S} \left(1 - \frac{m_i}{M_i}\right) M_i^2 \frac{s_i^2}{m_i} \\ &= 580^2 \left(1 - \frac{12}{580}\right) \frac{2611.8788}{12} + \frac{580}{12} \sum_{i \in S} \left(1 - \frac{3}{24}\right) (24)^2 \frac{s_i^2}{3} \\ &= 71,704,778.99 + 441,186.67 = 72,145,965.66 \end{aligned}$$

$\hat{t} = 50,653.3333, \quad \text{SE}(\hat{t}) = \sqrt{72,145,965.66} = 8493.88$
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Part (d): Estimate the mean number of worm fragments per can

From the lecture notes, we use the mean estimator:

$$\hat{\bar{y}} = \frac{1}{M_0} \hat{t}$$

Known total number of cans in the population:

$$M_0 = NM = 580 \times 24 = 13,920$$

Estimated mean:

$$\hat{\bar{y}} = \frac{50,653.3333}{13,920} = \mathbf{3.6389}$$

Standard error. The estimated variance of the mean is:

$$\hat{V}(\hat{y}) = \frac{\hat{V}(\hat{t})}{M_0^2} = \frac{72,145,965.66}{13,920^2} = 0.3723$$

$$\hat{y} = 3.6389, \quad \text{SE}(\hat{y}) = \sqrt{0.3723} = 0.6102$$

Did we need the ratio estimator? No. Because all cases contain the same number of cans ($M_i = M = 24$), $M_0 = NM$ is a known constant and the simple estimator $\hat{y} = \hat{t}/M_0$ is unbiased. The ratio estimator is only required when cluster sizes M_i vary.

Part (e): Verification using the survey package

```
library(survey)

corn <- data.frame(
  case_id = rep(1:12, each = 3),
  can_id  = rep(1:3, times = 12),
  worms   = c(
    1, 5, 7,
    4, 2, 4,
    0, 1, 2,
    3, 6, 6,
    4, 9, 8,
    0, 7, 3,
    5, 5, 1,
    3, 0, 2,
    7, 3, 5,
    3, 1, 4,
    4, 7, 9,
    0, 0, 0
  ),
  N_cases = 580,
  M_cans  = 24
)

corn_design <- svydesign(
  id   = ~ case_id + can_id,
  fpc  = ~ N_cases + M_cans,
  data = corn,
  nest = TRUE
)

# Estimated total
svytotal(~worms, corn_design)
```

```
##      total      SE
## worms 50653 8493.9
```

```
SE(svytotal(~worms, corn_design))
```

```
##           worms  
## worms 8493.878
```

```
# Estimated mean
```

```
svymean(~worms, corn_design)
```

```
##           mean      SE  
## worms 3.6389 0.6102
```

```
SE(svymean(~worms, corn_design))
```

```
##           worms  
## worms 0.6101924
```

The `survey` package confirms: total $\approx 50,653$ with SE ≈ 8493.9 , and mean ≈ 3.6389 with SE ≈ 0.6102 , matching parts (c) and (d).